

Beyond the Answer

Probing Numerical Reasoning in Financial LLMs

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GitHub Repository

github.com/aayushiiitd2025-byte/AAYUSH_LLM

Overview

This project investigates numerical reasoning failures in Financial Large Language Models (LLMs) using the **FinQA** dataset.

Rather than focusing only on final answer accuracy, this work studies:

- Retrieval failures
- Program reasoning correctness
- Context sensitivity
- Silent numerical reasoning failures
- Symbolic verification for reliability improvement

Configuration Used

- **Model:** Qwen2.5-0.5B-Instruct
 - **Environment:** Google Colab (T4 GPU)
 - **Dataset:** FinQA (8,281 QA pairs over S&P 500 earnings reports)
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Repository Structure

```
FinQA_Complete_Submission_Aayush.ipynb
NUMERICAL_REASONING_IN_FINANCIAL_LLMS_Aayush.pdf
AAYUSH_Q_AND_A.pdf
README.md
assets/
```

Layer 1 — Dataset Understanding

Performed detailed analysis of:

- FinQA symbolic program structure
- #N intermediate references
- Evidence source distribution
- Percentage reasoning patterns
- Execution vs Program accuracy

Key Findings

- Most questions depend heavily on table retrieval.
 - Execution accuracy alone is insufficient for financial reliability.
 - Program correctness is critical in regulated financial settings.
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Layer 2 — Error Taxonomy & Evaluation

Built a reasoning pipeline using **Qwen2.5-0.5B-Instruct**.

Evaluation Setup

- 100 FinQA test examples
- Few-shot prompting
- Execution accuracy evaluation
- Program accuracy comparison

Error Categories Analyzed

- Retrieval errors
- Operation/order errors
- Hallucinations
- Parsing failures
- Program mismatch

Major Findings

- Retrieval errors dominated (~93% of failures)
 - Few-shot prompting improved parse consistency
 - Execution accuracy and program accuracy frequently diverged
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Layer 2B — Context Sensitivity

Compared three context formats:

1. Full Context (Table + Text)
2. Table Only
3. Text Only

Observation

- Tables are the most critical information source for FinQA reasoning.
 - Removing tables significantly impacts numerical retrieval quality.
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Layer 3 — Proposed Improvement

Implemented:

- Complexity-based routing
- Symbolic verification layer

Pipeline

1. Classify question complexity
2. Route questions using different prompting strategies
3. Verify generated reasoning programs symbolically

Motivation

Financial systems require **reasoning reliability**, not merely plausible numerical outputs.

Result

Symbolic verification helped detect answer-program inconsistencies at zero additional inference cost.

Key Insight

A financial model can produce:

- a numerically plausible answer,
- with incorrect reasoning,
- without any visible indication of failure.

This creates a serious reliability issue in real-world financial decision-making systems. The goal of this project is therefore to improve:

- interpretability,
 - reasoning reliability,
 - verification transparency.
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Requirements

```
pip install transformers accelerate torch pandas matplotlib
```

Running the Notebook

```
FinQA_Complete_Submission_Aayush.ipynb
```

Run all cells sequentially in Google Colab.

Recommended Hardware

- Google Colab T4 GPU
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Files Included

- Complete implementation notebook
 - Layer 1 & Layer 2 written answers
 - Final presentation slides
 - Evaluation and analysis outputs
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Future Work

If additional compute resources were available:

- Fine-tune a domain-specific reasoning model
 - Add retrieval augmentation
 - Improve table structure encoding
 - Train a dedicated verifier model
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Acknowledgements

Dataset: FinQA — Numerical Reasoning over Financial Data

Model: Qwen2.5-0.5B-Instruct